

Covariance (S_{xy}) derivation

①

Show that

$$S_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \text{ can be written as}$$

$$S_{xy} = \frac{1}{n-1} \left(\sum_{i=1}^n x_i y_i - \frac{1}{n} \left[\sum_{i=1}^n x_i \sum_{i=1}^n y_i \right] \right)$$

Proof:

$$S_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n-1} \sum_{i=1}^n (x_i y_i - \bar{x} y_i - \bar{y} x_i + \bar{x} \bar{y})$$

$$= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \bar{x} \sum_{i=1}^n y_i - \bar{y} \sum_{i=1}^n x_i + n \bar{x} \bar{y} \right]$$

$$= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \bar{x} (n \bar{y}) - \bar{y} (n \bar{x}) + n \bar{x} \bar{y} \right]$$

$$= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y} \right]$$

$$= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - n \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \left(\frac{1}{n} \sum_{i=1}^n y_i \right) \right]$$

$$= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \frac{1}{n} \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right) \right]$$

②

Using S_{xy} , S_x and S_y we obtain the correlation coefficient:

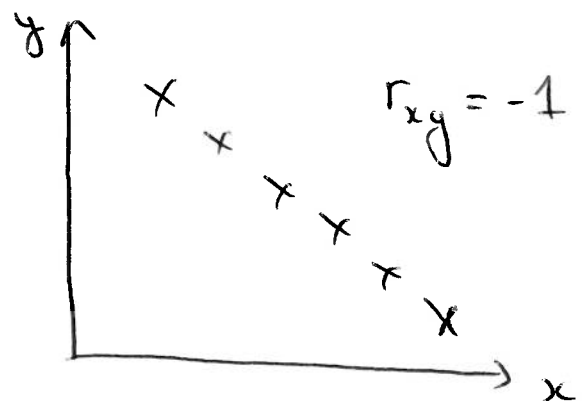
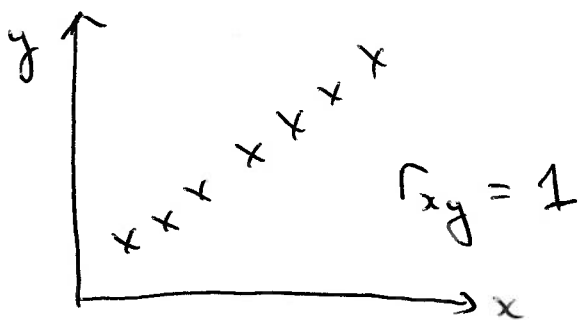
$$r_{xy} = \frac{S_{xy}}{S_x S_y}$$

and thus $S_{xy} = r_{xy} \times S_x \times S_y$

Note: $-1 \leq r_{xy} \leq 1$

$$S_{xy} \geq 0 \quad \text{or} \quad S_{xy} < 0$$

Examples:



Consider the data and calculate

$$\bar{x}, \bar{y}, S_x, S_y, r_{xy}, S_{xy}, \sum_{i=1}^n x_i, \sum_{i=1}^n y_i$$

$\sum_{i=1}^n x_i y_i$ with your calculator.

<u>X</u>	<u>Y</u>
1.2	50
1.8	75.2
1.25	42
1.72	72.5
1.84	90.3
1.99	70

① Use **INPUT** and **$\Sigma+$** to input the bivariate data

② Use **K SWAP** to move from X to Y variable.

Solutions

④

$$\bar{x} = 1.6333$$

$$\bar{y} = 66.6667$$

$$S_x = 0.3286$$

$$S_y = 17.6789$$

$$r_{xy} = 0.8496$$

$$S_{xy} = r_{xy} \times S_x \times S_y = 4.9356$$

$$\sum_{i=1}^n x_i = 9.8$$

$$\sum_{i=1}^n y_i = 400$$

$$\sum_{i=1}^n x_i y_i = 678.012$$